

Part A: Negative Number Operations*Evaluate each expression completely.*

1. $1 + (-5)$

2. $-7 + 8$

3. $-8 - 9$

4. $-7 - 12$

5. $11 + 4$

6. $-3 + (-4)$

7. $-8 - (-5)$

8. $7 \cdot (-4)$

9. $-72 \div (-9)$

10. $-9 \cdot 2$

11. $-9 \cdot (-3)$

12. $-3 \cdot (-4)$

13. $-10 \div 2$

14. $-42 \div (-6)$

15. $5 + (-9) \cdot 9$

16. $-63 \div (-7) - (-6)$

17. $-20 \div (-5) - (-18)$

18. $-72 \div 12 - (-18)$

19. $-13 + (-10) \cdot 12$

20. $-9 + 5 \cdot (-10)$

21. $(-2 + 10)^2 \div 4$

22. $-5 - 5 \cdot (7 + (-5))$

23. $(-7 + 2)^2 \div (-5)$

24. $(-8 + 7)^2 \div (-1)$

25. $9^3 - (-7) \cdot 4$

Part B: Solving Linear Equations

Solve each equation for x . Show your steps.

1. $x + 5 = -4$

2. $9x = -45$

3. $x - 6 = -11$

4. $4x = 28$

5. $5x - 3 = 17$

6. $6x - 3 = -57$

7. $5x + 9 = 44$

8. $6x + 8 = 14$

9. $4(x + 11) - 1 = 47$

10. $8(9x - 8) + 5 = 85$

11. $2(8x + 12) - 7 = -111$

12. $2(3x - 7) - 8 = 20$

13. $-9x - 11 = 5x - 53$

14. $4x + 3 = 5x + 5$

15. $-11x - 6 = -5x - 24$

16. $4x + 5 = 2x + 1$

17. $\frac{1}{3}x + 7 = 3$

18. $-\frac{1}{4}x + 2 = 1$

19. $\frac{x}{4} - 5 = -4$

20. $\frac{x}{4} - 6 = -\frac{7}{2}$

21. $\frac{x}{3} - 2 = -\frac{5}{3}$

$$22. \frac{5x + 40}{2} = \frac{2x - 10}{6}$$

$$23. \frac{6x - 76}{2} = \frac{-4}{2}$$

$$24. \frac{2x - 1}{3} = \frac{3}{3}$$

$$25. \frac{3x - 26}{4} = \frac{3}{12}$$

Part C: Exponent Rules

Simplify. Write answers with positive exponents.

$$1. 8x^8 \cdot 5x^7$$

$$2. 3x^7 \cdot 7x^7$$

$$3. 10x^2 \cdot 9x^7$$

$$4. 8x^9 \cdot 7x^2$$

$$5. \frac{6x^{20}}{x^{10}}$$

$$6. \frac{3x^{12}}{x^{10}}$$

7. $\frac{3x^7}{x^5}$

8. $\frac{2x^{11}}{x^6}$

9. $(4x^4)^4$

10. $(8x^8)^4$

11. $(7x^4)^5$

12. $(2x^3)^4$

13. $7 \cdot 10^{-2}$

14. $\frac{2}{4^{-1}}$

15. $\frac{2}{2^{-2}}$

16. $8 \cdot 10^{-1}$

17. $5^5 \cdot 5^{-2}$

18. $7^3 \cdot 7^{-4}$

19. $3^4 \cdot 3^{-4}$

20. $3^3 \cdot 3^{-2}$

21. $(6x^{-1})^3 \cdot x^{10}$

22. $\frac{6x^4y^4}{3x^3y^2}$

23. $(10x^{10}y^4)^3 \cdot 5x^9y^2$

24. $\left(\frac{5x^{11}}{5x^6}\right)^2$

25. $\frac{7x^{14}y^{10}}{x^4y^7}$

Part D: Sale Price and Tax

Solve each word problem. Show your work.

1. A guitar that normally costs \$25 is on sale for 12% off. Find the sale price.
2. A tablet that normally costs \$60 is on sale for 5% off. Find the sale price.
3. A guitar that normally costs \$40 is on sale for 60% off. Find the sale price.
4. A store marks up a backpack that costs \$70 by 30%. Find the new price.

5. A backpack that normally costs \$220 is on sale for 40% off. Find the sale price.
6. A jacket that normally costs \$55 is on sale for 60% off. Find the sale price.
7. A guitar that normally costs \$225 is on sale for 12% off. Find the sale price.
8. A store marks up a skateboard that costs \$10 by 40%. Find the new price.
9. A store marks up a bicycle that costs \$50 by 18%. Find the new price.
10. A skateboard that normally costs \$20 is on sale for 30% off. Find the sale price.
11. A guitar that normally costs \$260 is on sale for 15% off. Find the sale price.
12. A skateboard that normally costs \$240 is on sale for 30% off. Find the sale price.
13. A tablet that normally costs \$65 is on sale for 60% off. Find the sale price.
14. Maya's bill comes to \$80. Including a 60% tip, find the total amount Maya pays.

15. Ruby's bill comes to \$340. Including a 5% tip, find the total amount Ruby pays.
16. Leo's bill comes to \$210. Including a 30% sales tax, find the total amount Leo pays.
17. Mia's bill comes to \$20. Including a 30% sales tax, find the total amount Mia pays.
18. Mia's bill comes to \$30. Including a 30% sales tax, find the total amount Mia pays.
19. Ruby's bill comes to \$305. Including a 40% sales tax, find the total amount Ruby pays.
20. Jonas's bill comes to \$20. Including a 15% tip, find the total amount Jonas pays.
21. Liam's bill comes to \$50. Including a 18% sales tax, find the total amount Liam pays.
22. Nina's bill comes to \$165. Including a 20% sales tax, find the total amount Nina pays.
23. Leo's bill comes to \$40. Including a 5% sales tax, find the total amount Leo pays.
24. Diego's bill comes to \$44. Including a 25% sales tax, find the total amount Diego pays.

25. Ava's bill comes to \$44. Including a 25% sales tax, find the total amount Ava pays.

Answer Key

Part A: Negative Number Operations

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|----------|-----------|------------|
| 1. -4 | 10. -18 | 19. -133 |
| 2. 1 | 11. 27 | 20. -59 |
| 3. -17 | 12. 12 | 21. 16 |
| 4. -19 | 13. -5 | 22. -15 |
| 5. 15 | 14. 7 | 23. -5 |
| 6. -7 | 15. -76 | 24. -1 |
| 7. -3 | 16. 15 | 25. 757 |
| 8. -28 | 17. 22 | |
| 9. 8 | 18. 12 | |

Part B: Solving Linear Equations

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|-------------|---------------|---------------|
| 1. $x = -9$ | 10. $x = 2$ | 19. $x = 4$ |
| 2. $x = -5$ | 11. $x = -8$ | 20. $x = 10$ |
| 3. $x = -5$ | 12. $x = 7$ | 21. $x = 1$ |
| 4. $x = 7$ | 13. $x = 3$ | 22. $x = -10$ |
| 5. $x = 4$ | 14. $x = -2$ | 23. $x = 12$ |
| 6. $x = -9$ | 15. $x = 3$ | 24. $x = 2$ |
| 7. $x = 7$ | 16. $x = -2$ | 25. $x = 9$ |
| 8. $x = 1$ | 17. $x = -12$ | |
| 9. $x = 1$ | 18. $x = 4$ | |

Part C: Exponent Rules

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|----------------|---------------------|------------------------|
| 1. $40x^{15}$ | 10. $4096x^{32}$ | 19. 1 |
| 2. $21x^{14}$ | 11. $16807x^{20}$ | 20. 3 |
| 3. $90x^9$ | 12. $16x^{12}$ | 21. $216x^7$ |
| 4. $56x^{11}$ | 13. $\frac{7}{100}$ | 22. $2xy^2$ |
| 5. $6x^{10}$ | 14. 8 | 23. $5000x^{39}y^{14}$ |
| 6. $3x^2$ | 15. 8 | 24. x^{10} |
| 7. $3x^2$ | 16. $\frac{4}{5}$ | 25. $7x^{10}y^3$ |
| 8. $2x^5$ | 17. 125 | |
| 9. $256x^{16}$ | 18. $\frac{1}{7}$ | |

Part D: Sale Price and Tax

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|----------------|-----------------|-----------------|
| 1. 22 dollars | 10. 14 dollars | 19. 427 dollars |
| 2. 57 dollars | 11. 221 dollars | 20. 23 dollars |
| 3. 16 dollars | 12. 168 dollars | 21. 59 dollars |
| 4. 91 dollars | 13. 26 dollars | 22. 198 dollars |
| 5. 132 dollars | 14. 128 dollars | 23. 42 dollars |
| 6. 22 dollars | 15. 357 dollars | 24. 55 dollars |
| 7. 198 dollars | 16. 273 dollars | 25. 55 dollars |
| 8. 14 dollars | 17. 26 dollars | |
| 9. 59 dollars | 18. 39 dollars | |